

```

noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:15
●) g++ lab-1.cpp -o ./build/lab-1 && ./build/lab-1
Enter 1 to encrypt or 2 to decrypt: 1
Enter text (lowercase): top secret message
Enter Key value: 5
Encrypted: ytu xjhwjy rjxxflj

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:16
●) ./build/lab-1
Enter 1 to encrypt or 2 to decrypt: 2
Enter text (lowercase): ytu xjhwjy rjxxflj
Enter Key value: 5
Decrypted: top secret message

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:20
●) g++ lab-2.cpp -o ./build/lab-2 && ./build/lab-2
Enter 1 to encrypt or 2 to decrypt: 1
Enter text (lowercase): top secret message
Enter substitution key (26 chars, a-z): qwertyuiopasdfghjklzxcvbnm
Encrypted: zgh ltektz dtllqut

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:21
●) ./build/lab-2
Enter 1 to encrypt or 2 to decrypt: 2
Enter text (lowercase): zgh ltektz dtllqut
Enter substitution key (26 chars, a-z): qwertyuiopasdfghjklzxcvbnm
Decrypted: top secret message

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:31
●) g++ lab-3.cpp -o ./build/lab-3 && ./build/lab-3
Enter 1 to encrypt or 2 to decrypt: 1
Enter text: manoftheword
Enter key: monarchy
Encrypted: orankpcfvdnk

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:31
●) ./build/lab-3
Enter 1 to encrypt or 2 to decrypt: 2
Enter text: orankpcfvdnk
Enter key: monarchy
Decrypted: manoftheword

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:48
●) g++ lab-4.cpp -o ./build/lab-4 && ./build/lab-4
Enter 1 for encryption and 2 for decryption: 1
===== HILL CIPHER ALGORITHM =====

Key Matrix:
3 3
2 5

Enter plaintext: IAMIRONMAN

Encrypted text: YQIMPAXINN

Programmed by Milan Bairagi...
noname ~/bit-labs/5th Semester/Information Technology % main ! v15.2.0 14:49
●) g++ lab-4.cpp -o ./build/lab-4 && ./build/lab-4
Enter 1 for encryption and 2 for decryption: 2
===== HILL CIPHER ALGORITHM =====

Key Matrix:
3 3
2 5

Enter ciphertext: YQIMPAXINN

Decrypted text: IAMIRONMAN

Programmed by Milan Bairagi...

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noname ~/bit-labs % main ! 15:20
●) cd "/home/noname/bit-labs/5th Semester/Information Technology/" && mkdir -p build && g++ lab-5.cpp -o ./build/lab-5 && ./build/lab-5
===== VIGENERE CIPHER ALGORITHM =====

Enter the plaintext: NONAME
Enter the key: WHO

Plaintext: NONAME
Encrypted text: JVBWTS

Ciphertext: JVBWTS
Decrypted text: NONAME

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 16:05
$ g++ lab-6.cpp -o ./build/lab-6 && ./build/lab-6
===== VERNAM CIPHER ALGORITHM =====

Enter the plaintext: SECRETMESAGE

Generated key: UPIQMTQAMJTDD

Plaintext: SECRETMESAGE
Encrypted text: MTKHQMCEEETJH

Ciphertext: MTKHQMCEEETJH
Decrypted text: SECRETMESAGE

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 13:51
$ g++ lab-7.cpp -o ./build/lab-7 && ./build/lab-7

===== Rail Fence Cipher Algorithm =====
Enter plaintext: HELLO WORLD
Enter number of rails: 3

===== ENCRYPTION =====
Plaintext: HELLO WORLD
Number of rails: 3
Encrypted text: HOREL OLLWD

===== DECRYPTION =====
Ciphertext: HOREL OLLWD
Number of rails: 3
Decrypted text: HELLO WORLD

Programmed by Milan Bairagi...

```

```

noname ~/bit-labs % main ! 14:17
$ cd "/home/noname/bit-labs/5th Semester/Information Security/" && mkdir -p build && g++ lab-8.cpp -o ./build/lab-8 && ./build/lab-8

===== Euclidean Algorithm =====
Enter first number: 48
Enter second number: 18

Euclidean Algorithm Steps:
=====
Step | Dividend | Divisor | Remainder | Quotient
-----|-----|-----|-----|-----
1 | 48 | 18 | 12 | 2
2 | 18 | 12 | 6 | 1
3 | 12 | 6 | 0 | 2
=====

GCD(48, 18) = 6

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 14:43
$ g++ lab-9.cpp -o ./build/lab-9 && ./build/lab-9

===== First 20 Prime Numbers =====
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71

Programmed by Milan Bairagi...

```

```

noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 15:05
$ g++ lab-10.cpp -o ./build/lab-10 && ./build/lab-10
Enter a number: 19
19 is prime

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 16:03
$ g++ lab-11.cpp -o ./build/lab-11 && ./build/lab-11
=== Euler's Theorem: Checking Relatively Prime Numbers ===

Enter first number: 9
Enter second number: 11

=== Results ===
9 and 11 are relatively prime.
GCD(9, 11) = 1

=== Verifying Euler's Theorem ===
Euler's Totient  $\phi(11) = 10$ 
 $9^{10} \bmod 11 = 1$ 
Euler's Theorem verified:  $a^{\phi(n)} \equiv 1 \pmod{n}$ 

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 16:25
> g++ lab-12.cpp -o ./build/lab-12 && ./build/lab-12
Enter a number: 11
Numbers coprime to 11: 1 2 3 4 5 6 7 8 9 10
Euler's Totient Function for 11 is 10

Programmed by Milan Bairagi...

```

```

noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 17:20
> g++ lab-13.cpp -o ./build/lab-13 && ./build/lab-13
=== Diffie-Hellman Key Exchange Algorithm ===
Public Key Generation:
Prime number (P) = 23
Primitive root (G) = 5

Alice's private key (a) = 4
Bob's private key (b) = 3

Alice's public key (x) =  $G^a \bmod P = 4$ 
Bob's public key (y) =  $G^b \bmod P = 10$ 

Alice(x) and Bob(y) exchange their public keys.

Shared secret (Alice):  $y^a \bmod P = 18$ 
Shared secret (Bob):  $x^b \bmod P = 18$ 

Programmed by Milan Bairagi...

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noname ~/bit-labs/5th Semester/Information Security % main ! v15.2.0 17:25
> g++ lab-14.cpp -o ./build/lab-14 && ./build/lab-14
=== RSA Algorithm ===

=== Key Generation ===
Chosen prime numbers:
p = 7919
q = 1009
n = p * q = 7919 * 1009 = 7990271
 $\phi(n) = (p-1)(q-1) = 7918 * 1008 = 7981344$ 

Choose public exponent (e) where  $1 < e < \phi(n)$ :
Public exponent (e) = 5
Private exponent (d) = 1596269 (where  $e * d \equiv 1 \pmod{\phi(n)}$ )

=== Public and Private Keys ===
Public Key (e, n) = (5, 7990271)
Private Key (d, n) = (1596269, 7990271)

=== Encryption/Decryption ===
Plaintext message (M) = 123

Encryption:
Ciphertext (C) =  $M^e \bmod n = 123^5 \bmod 7990271 = 3332110$ 

Decryption:
Decrypted Message (M) =  $C^d \bmod n = 3332110^{1596269} \bmod 7990271 = 123$ 

Programmed by Milan Bairagi...

```



Tools About

Des Encryption / Decryption Tool

DES Encryption

Encryption Text

The top secret message.

Secret Key

password123

Encryption Mode

CBC ECB

IV (optional)

IV

Output format

Base64 HEX

Encrypt

Encrypted Text

sj07D6iby1/7PH8c3Ts45CabG/Kc2Z33

Encryption Tools

- AES Encryption / Decryption
- RSA Encryption / Decryption
- DES Encryption / Decryption
- Triple DES Encryption / Decryption
- Jasypt Encryption / Decryption

Encoding Tools

- Base64 Encoding / Decoding
- Hex Encoding / Decoding

Hashing Tools

- Bcrypt Hash Generator
- MD5 Hash Generator
- SHA 256 Hash Generator
- HMAC Hash Generator

Image Tools

- Image to PDF Converter
- Image Converter
- Image Size Reducer
- Video Thumbnail Generator

anycrypt.com/crypto/des

Encrypt

DES Encryption

Encrypted Text

sj07D6iby1/7PH8c3Ts45CabG/Kc2Z33

Secret Key

password123

Encryption Mode

CBC

ECB

IV (optional)

IV

Input format

Base64

HEX

Decrypt

Decrypted Text

The top secret message.

Image Size Reducer

Video Thumbnail Generator

Utility Tools

Date Difference Calculator

Age Calculator

Fuel to Distance Calculator

PDF Merger

Finance Calculators

Fixed Deposit Calculator

SIP Calculator

TIP Calculator

Unit Converters

Length Unit Converter

anycrypt.com/crypto

Tools About

AES Encryption / Decryption Tool

AES Encryption

Encryption Text

The top secret message.

Secret Key

password12345678

Encryption Key Size

128 Bits

192 Bits

256 Bits

Encryption Mode

CBC

ECB

IV (optional)

IV

Output format

Base64

HEX

Encrypt

Encrypted Text

CV77jXhdeW7rfrceVE95WtQ6LHxqe/bvrjULnO//
DNYM=

Encryption Tools

AES Encryption / Decryption

RSA Encryption / Decryption

DES Encryption / Decryption

Triple DES Encryption / Decryption

Jasypt Encryption / Decryption

Hashing Tools

Bcrypt Hash Generator

MD5 Hash Generator

SHA 256 Hash Generator

HMAC Hash Generator

Encoding Tools

Base64 Encoding / Decoding

Hex Encoding / Decoding

Image Tools

Image to PDF Converter

Image Converter

Image Size Reducer

Video Thumbnail Generator

Utility Tools

Date Difference Calculator

anycrypt.com/crypto

AES Decryption

Encrypted Text

CV77jXhdeW7rfrceVE95WtQ6LHxqe/bvrjULnO//
DNYM=

Secret Key

password12345678

Encryption Key Size

128 Bits

192 Bits

256 Bits

Encryption Mode

CBC

ECB

IV (optional)

IV

Input format

Base64

HEX

Decrypt

Decrypted Text

The top secret message.

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SHA256 Encrypt/Decrypt

SHA384 Encrypt/Decrypt

SHA256 Encrypt/Decrypt

Share

Add to Favs

Report Bug

Encrypter

Decrypter

Text

I am IronMan.

SHA256 Hash

a33bf115e612b6663e50e2d99296462ffa2d1d6906cbbd
f7f0004b11342370cf

Encrypt >

Reset

Copy